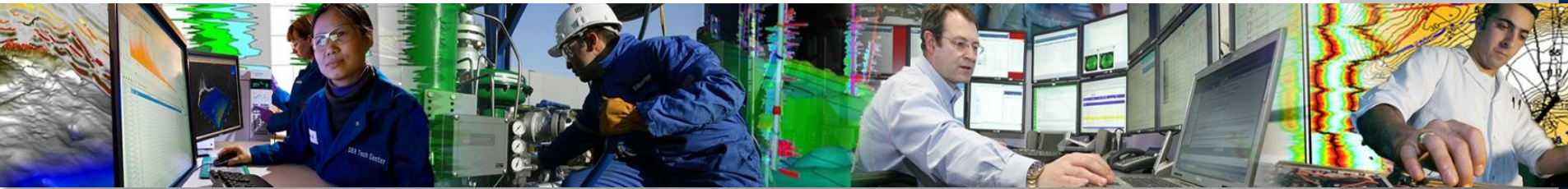


Mapping the Subsurface - with Petrel



Introduction to Fault Mapping

Key Concepts for Fault interpretation

1. To provide the most accurate and sound geologic interpretation in faulted areas; the interpretation, mapping, and validation of the faults is the first step.
2. An understanding of the regional tectonic setting will guide the interpretation of fault and horizon construction.
3. Horizon contouring and Fault plan contouring go hand-in-hand when defining structural relationships.

Tearpock and Bischke, 2003

Key Aspects of Fault Surface Maps

1. Aid in solving 3D problems.
2. Defines the location of a fault in space horizontally and vertically.
3. Aids in defining complex fault patterns.
4. Used to integrate Structure horizon maps to define position of key intervals from one side of a fault to another.
5. Used to define possible cross fault seal and drainage potential.
6. Aid in estimating dip, strike and displacement of a surface or horizon to any position in 3D space.

Fault Maps and Systems

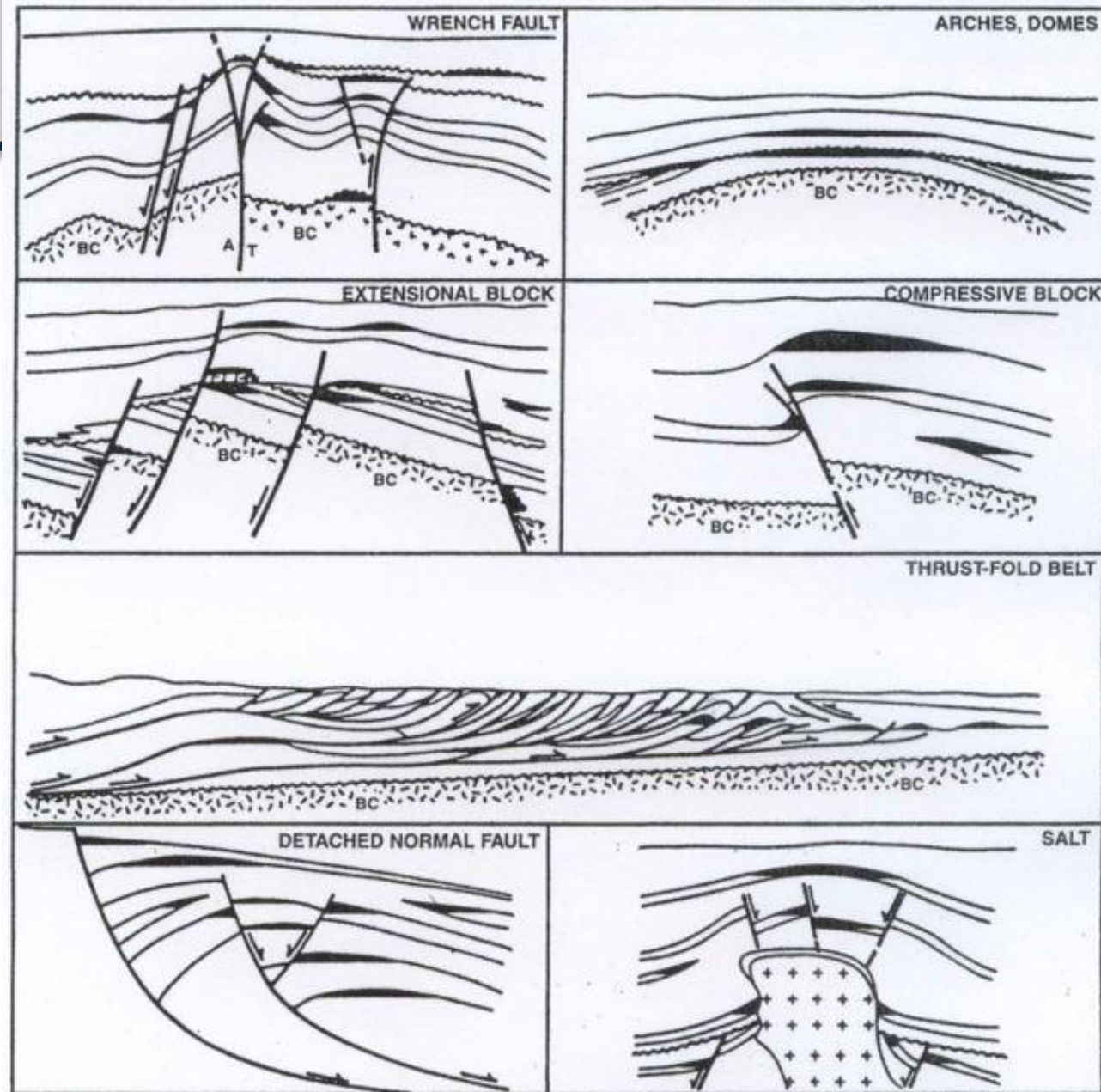
- Play a significant role in trapping hydrocarbons
- Explorationist need to understand fault origin and relationship to formation of structures
- Fault surface maps constructed from
 - Well log correlations
 - Interpreted seismic
 - outcrops

Fault Systems

- Extensional/normal
- Reverse/Compressional
- Growth/Detached Normal
- Wrench/Strike-slip

Structural Styles

- Specific assemblage of geologic structures in an area
- To prepare geologically reasonable maps, one must be familiar with the tectonic setting
 - Fault and structure type expected
 - Origin
 - Development



Fault Terminology

- Vertical separation

- The vertical component of bed displacement
 - Measured as the vertical distance between a horizon projected from one fault block across a fault to a point where the projection is vertically over or under the same horizon in the opposite fault block.
 - Seen in vertical wellbores or vertical cross sections

- Throw

- The difference in vertical depth between the fault intersection with a horizon in one fault block and the fault intersection with the same horizon in the opposing fault block
 - Determined in a direction perpendicular to the strike of the fault surface

- Heave

- The horizontal distance between the fault intersection with a horizon in one fault block and the fault intersection with the same horizon in the opposing fault block
 - Determined in a direction perpendicular to the strike of the fault surface

More fault terminology

- Missing section =
 - The vertical thickness of the stratigraphic section faulted out of a wellbore as a direct result of a normal fault cutting through the section
 - Missing section is referred to as fault cut
- Repeated section =
 - The vertical thickness of the stratigraphic section repeated in a wellbore as the direct result of a reverse fault cutting through the section
 - The repeated section can be determined by correlation of electric logs.

Technically, *missing* or *repeated* section is equal in value to vertical separation.

Missing or *repeated* section recognized in well logs is **NOT throw**.

- The terms **throw and vertical separation** are commonly misused in the preparation of subsurface interpretations and the accompanying fault and horizon maps

Fault Descriptive Terms

1. **Vertical separation (AE)** is the vertical component of bed displacement.
2. **Throw (AC)** determined in a direction perpendicular to the strike of the fault surface in the vertical direction to some horizon in one fault block compared to the other. (eq=AC or $AB \sin \theta$)
3. **Heave (BC)** determined in a direction perpendicular to the strike of the Fault in a horizontal direction to some horizon in one fault block compared to the other.
4. **Missing Section** is the vertical thickness of the stratigraphic section faulted out of a well bore or vertical profile as a direct result of the fault cutting the section
5. **Slip** describes the relative displacement of a formally joined point along a fault and is the apparent relative displacement along that surface.

Tearpock and Bischke, 2003

Block Diagram of a Fault

The block and the bed it contains, x, are offset at a normal fault.

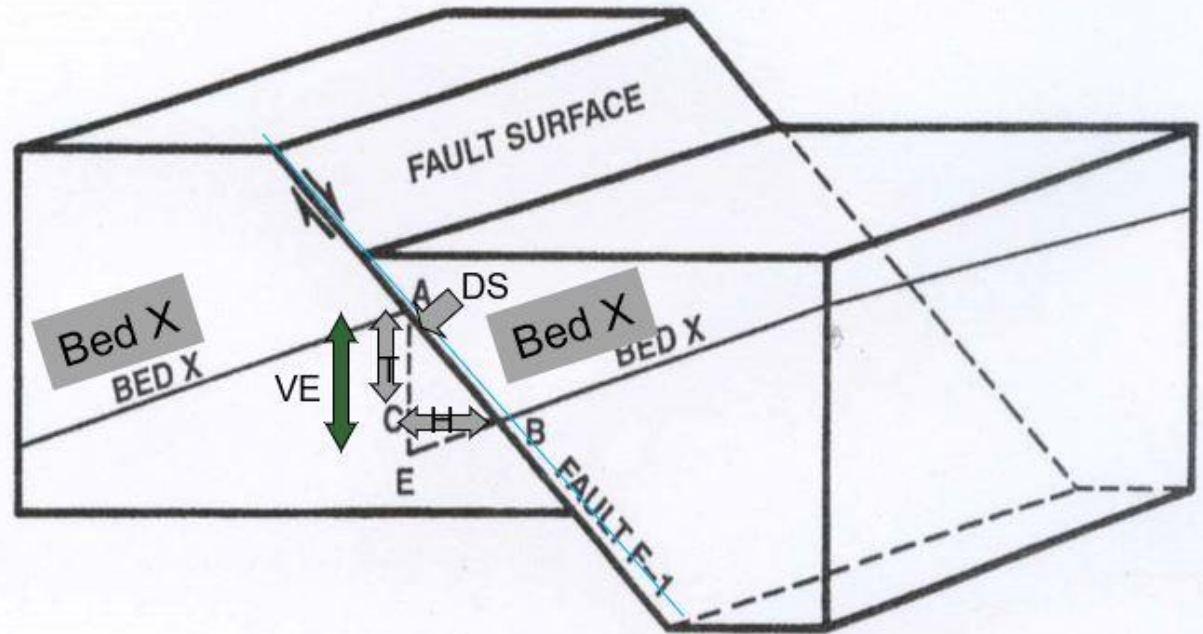
Four components of fault displacement are shown.

AB = dip slip

AC = throw

AE = Vertical separation

BC = Heave



BLOCK DIAGRAM

AB = DIP SLIP

AC = THROW

AE = VERTICAL SEPARATION

BC = HEAVE

Rules for contouring Fault Surfaces

1. Contours on faults maybe open-ended. Lateral termination.
2. Changes in dip-strike are assumed to be gradual.
3. Generally normal faults have smooth strike and dip curves contours unless interpreted by some other event.
4. Use the interpretive method of contouring outlined in Ch.2 for fault preparation.
5. Contours for several faults may map on a single base map and show merging affects that denote fault interaction.
6. Fault surface maps should represent the geologic regime being mapped.
7. Generally a 500- to 1000ft contour interval is used but this can be smaller based on the events being evaluated.

Rules for Contouring Fault Surfaces and Horizons

- The contours of a fault surface join those of a given mapping horizon at points of intersection of fault contours and horizon contours or equal value.
- Faults commonly dip at different angles than a key map horizon and consequently, only some of the fault surface contours intersect with those of a given datum
- The fault surface intersects horizons above and below a given horizon unless the fault is extremely limited in vertical extent.

Tearpock and Bischke, 2003

Fault Surface Contouring Guidelines

- Contours on a fault surface may be open-ended.
 - Faults may terminate laterally in the subsurface area of interest
- Changes in either fault strike or dip are assumed to be gradual
 - smooth as opposed to abrupt changes between linear segments
- The interpretive method of contouring is usually best
- Several fault surfaces may be contoured on a single base map
 - Where a fault overlies another the underlying contours are dashed
- Fault surface maps are usually mapped with a 500' or 1000' contour interval
 - Fault surfaces can be steep
 - Thrust faults may dip at low angles
 - A smaller CI may be appropriate for this case